Exemple d'utilisation du package levysim

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Préparation

Installation du package depuis GitHub (à faire une seule fois, puis redémarrer R) :

```
remotes::install_github("rytracetito/levy_packages", subdir = "levysim")
```

Tous les exemples utilisent les **processus homogènes** : les fonctions `sim_` sont appelées sans fonction du temps, et les valeurs par défaut (`A`, `Lambda` et `tau` égales à l'identité) donnent des processus de Lévy. Le processus inverse gamma n'a pas de version de Lévy ; on prend la fonction de forme linéaire $\eta(w) = w/\alpha$ (`beta = 1`), pour laquelle le processus gamma sous-jacent $T(w)$ est homogène en niveau.

Paramètres utilisés :

Processus	Fonction	Paramètres
Gamma	`sim_gamma`	`a = 2`, `b = 3` (taux)
Wiener	`sim_wiener`	`mu = 0.5`, `sigma = 1.5`
Inverse gaussien	`sim_ig`	`mu = 0.5`, `eta = 2`
Variance-gamma	`sim_vg`	`theta = 0.5`, `sigma = 1.5`, `nu = 2`
Meixner	`sim_meixner`	`a = 1`, `b = 0.3`, `d = 1.5`, `m = 0.2`
Inverse gamma	`sim_igam`	`theta = 1`, `alpha = 1`, `beta = 1`

Représentation d'un plan d'inspection :

```
tracer_plan <- function(t, titre) {
  ggplot(data.frame(t = t), aes(x = t, y = 0)) +
    geom_point() +
    labs(title = titre, x = "Temps", y = NULL) +
```

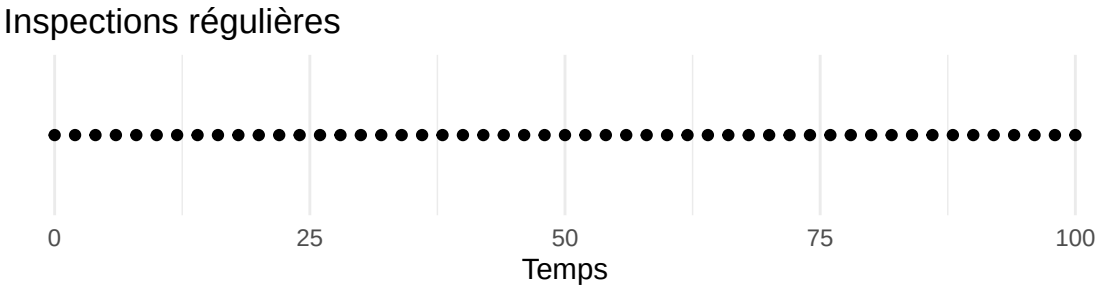
```
    theme_minimal() +
    theme(axis.text.y = element_blank(),
          panel.grid.major.y = element_blank(),
          panel.grid.minor.y = element_blank())
}
```

I Une trajectoire

Une trajectoire inspectée 50 fois entre 0 et 100.

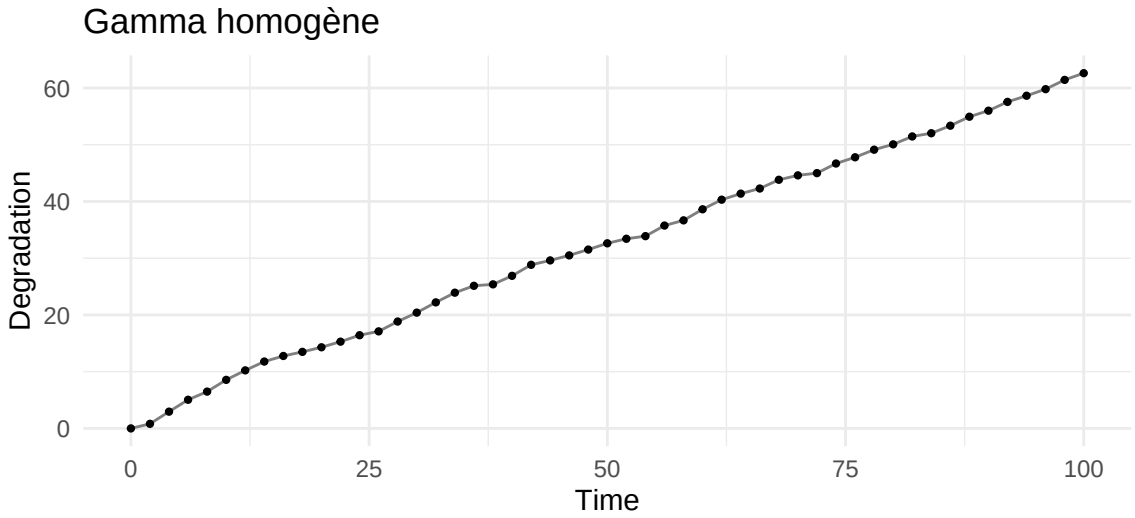
I.1 Cas 1 : inspections régulières

```
set.seed(1)
t1 <- inspection_times("regular", n = 50, tmax = 100)$time
tracer_plan(t1, "Inspections régulières")
```



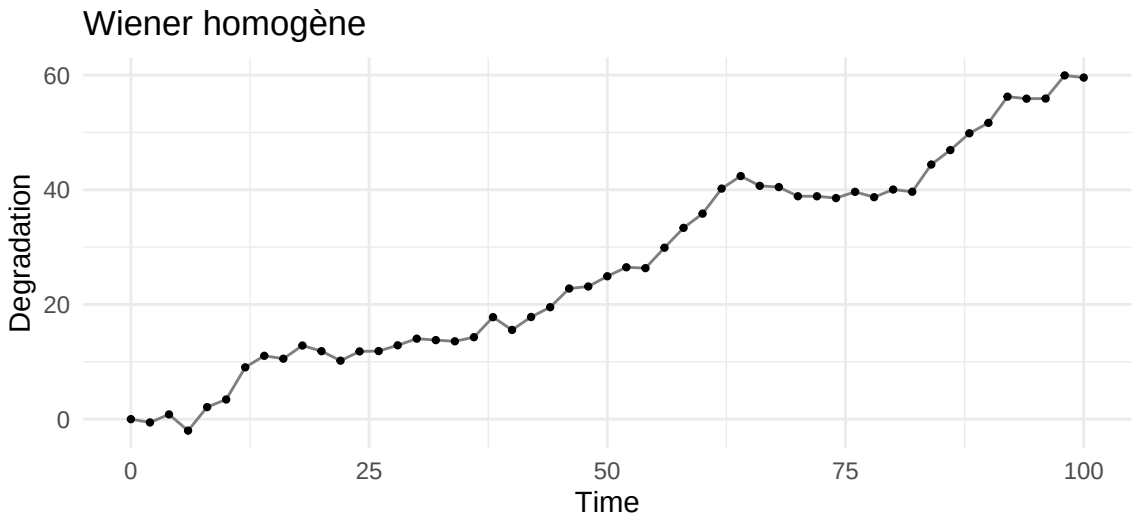
I.1.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t1, a = 2, b = 3), "Gamma homogène")
```



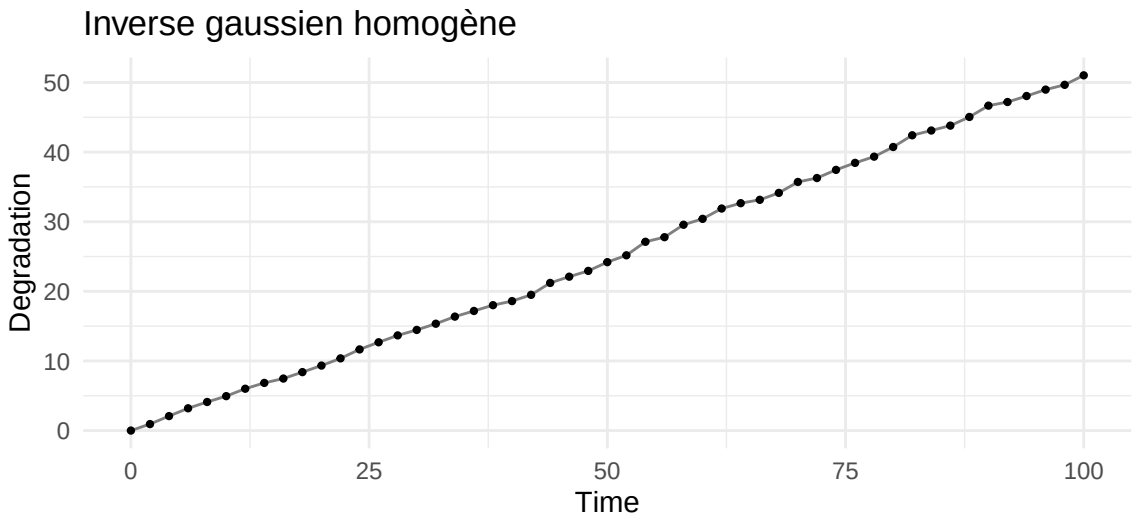
I.1.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t1, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



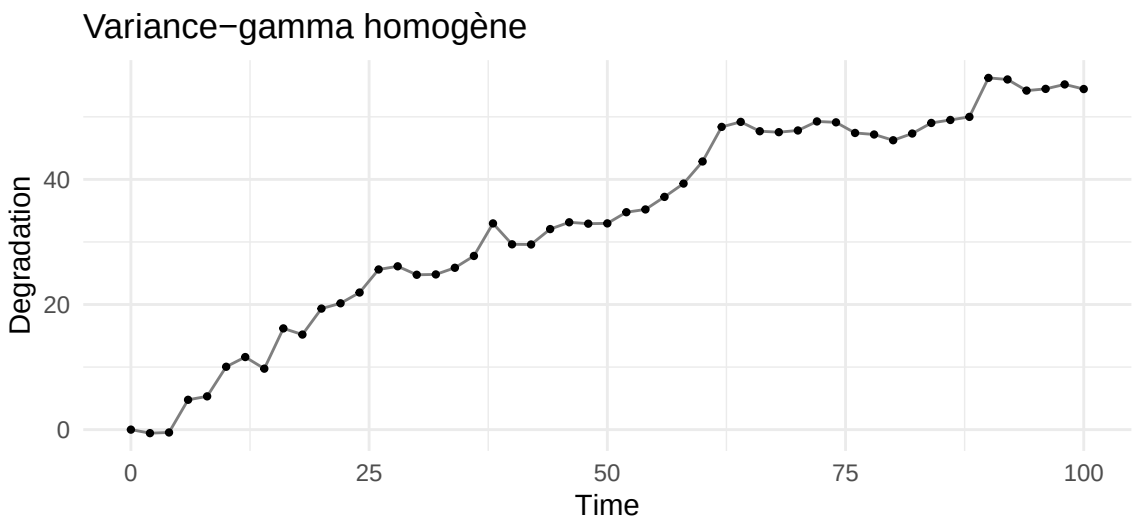
I.1.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t1, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



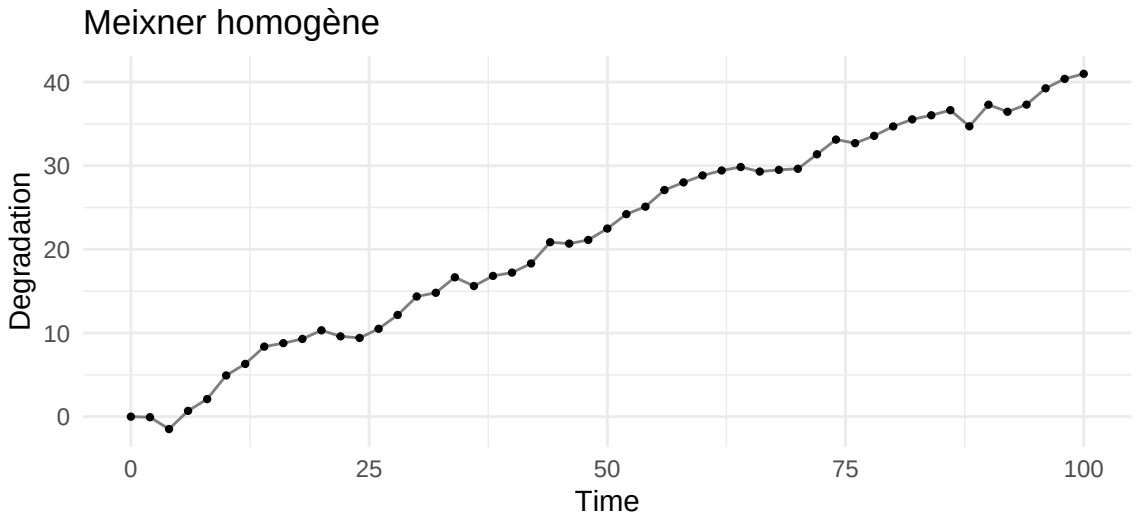
I.1.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t1, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



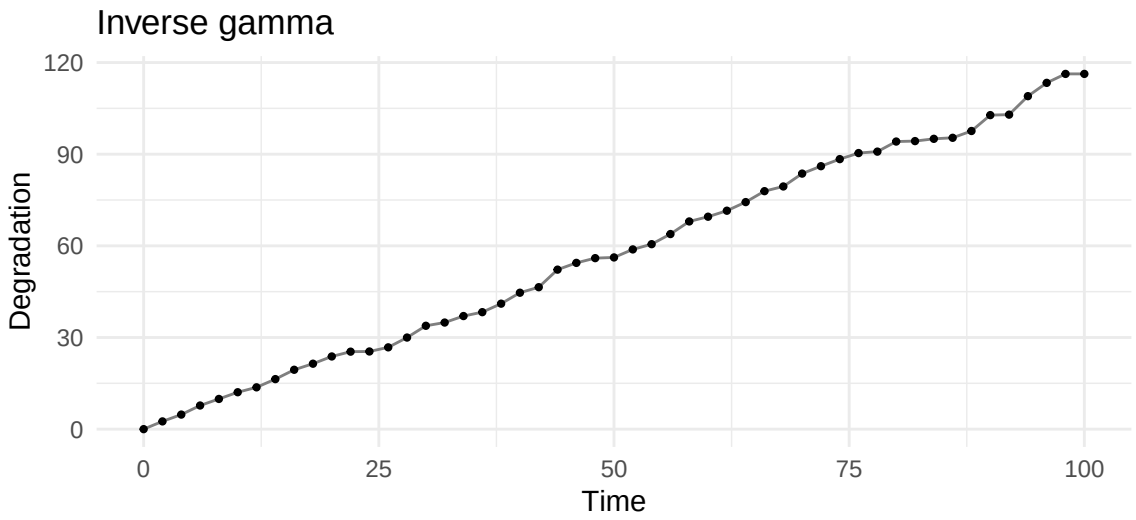
I.1.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(t1, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



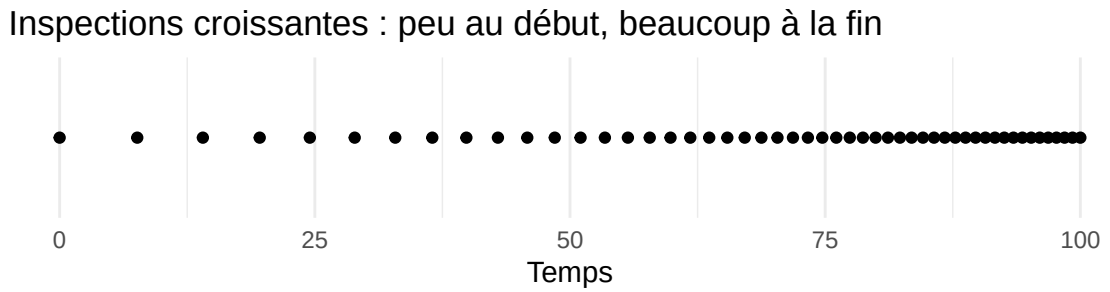
I.1.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(t1, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
Degradation
120
90
60
30
0
0
25
50
75
100
Time

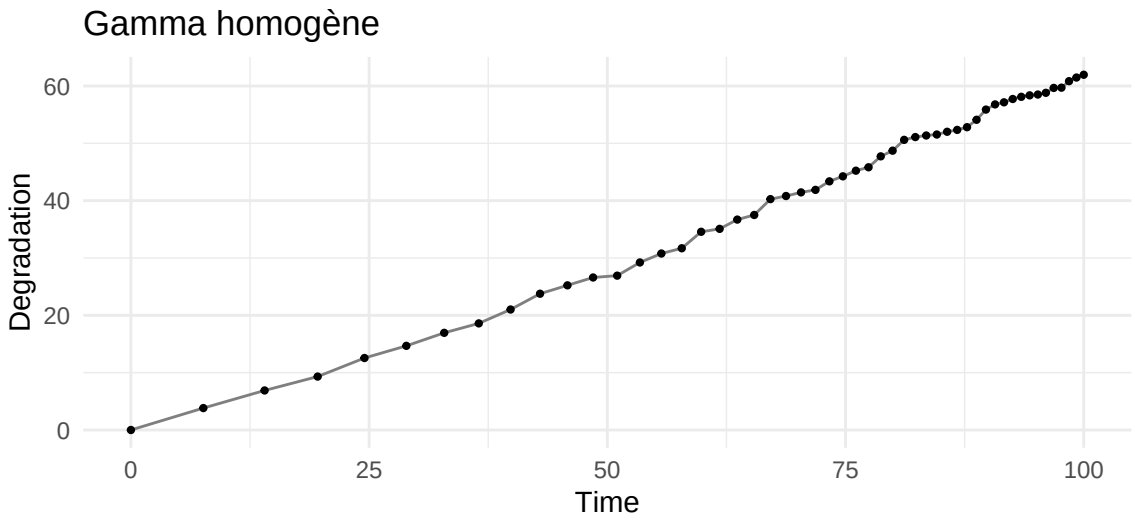
I.2 Cas 2 : inspections croissantes

```
set.seed(2)
t2 <- inspection_times("increasing", n = 50, tmax = 100)$time
tracer_plan(t2, "Inspections croissantes : peu au début, beaucoup à la fin")
```



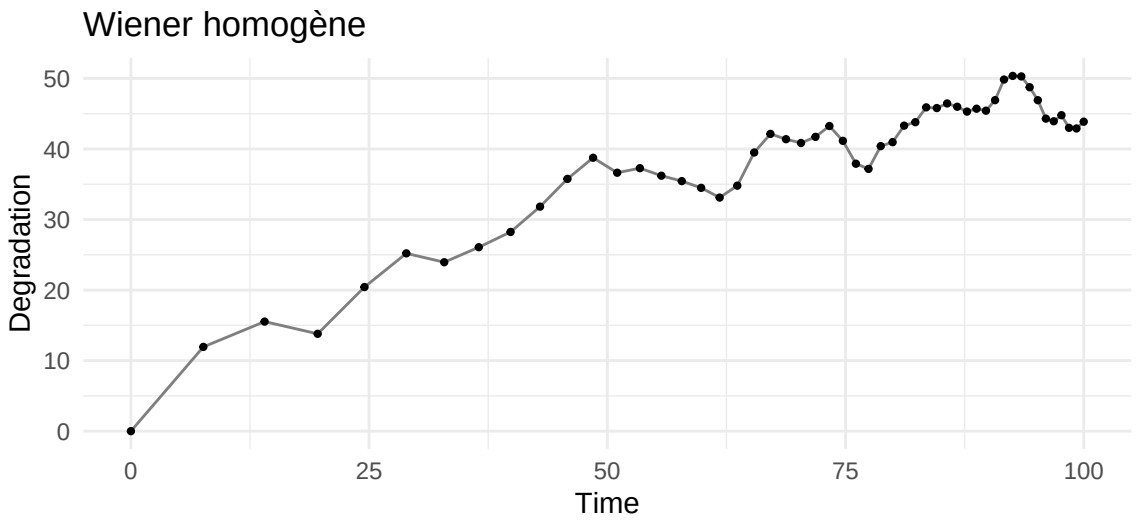
I.2.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t2, a = 2, b = 3), "Gamma homogène")
```



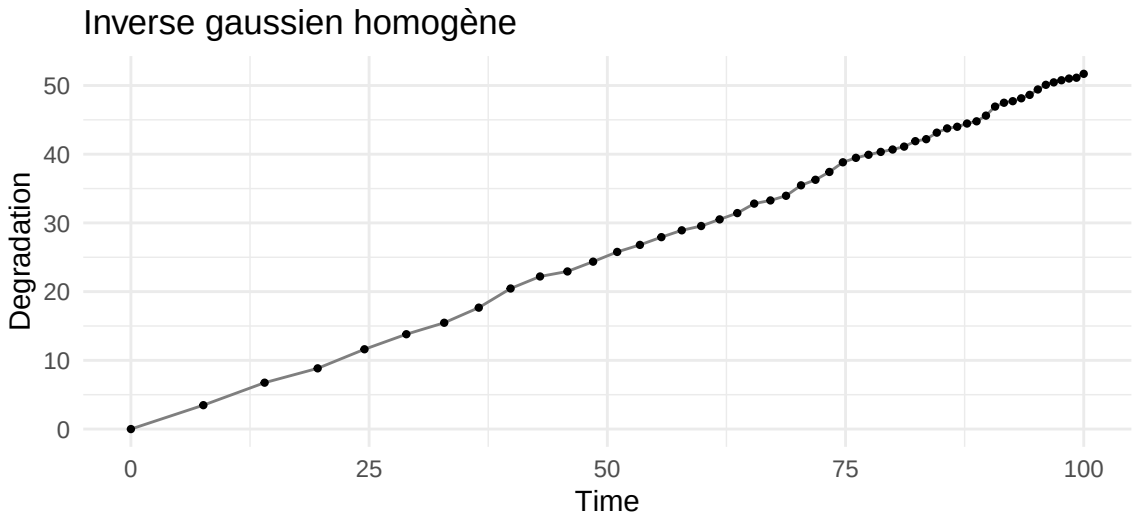
I.2.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t2, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



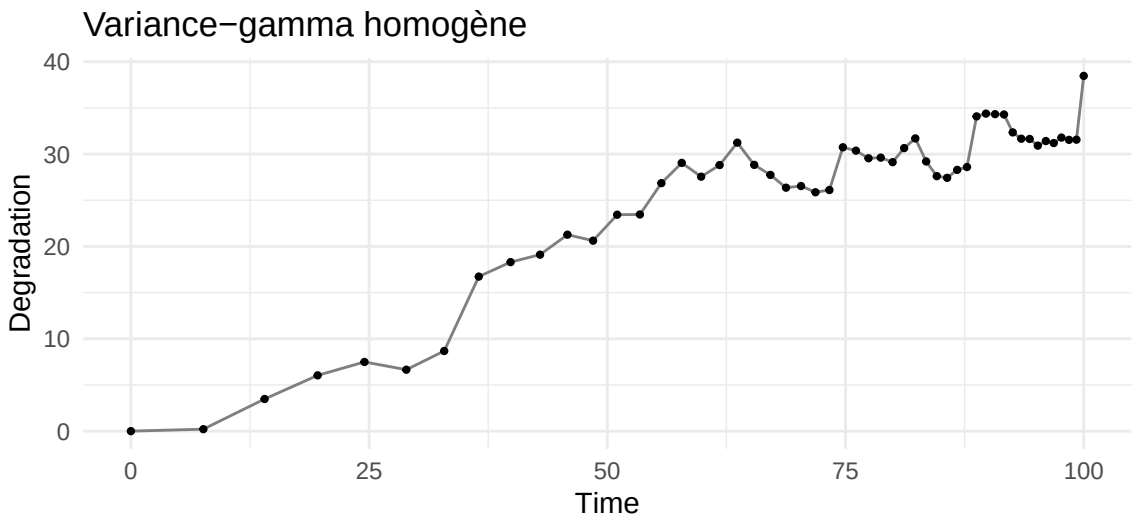
I.2.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t2, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



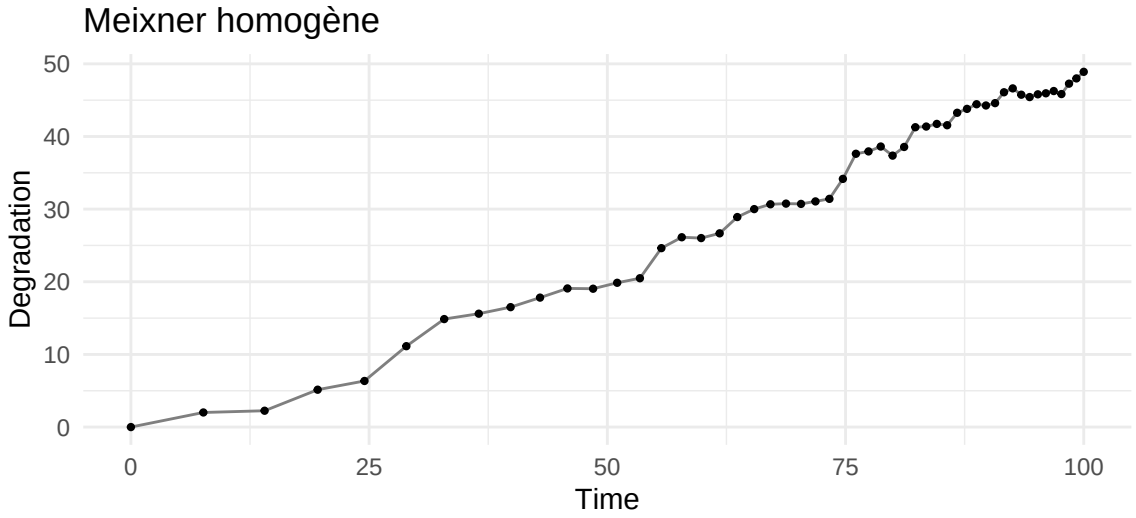
I.2.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t2, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



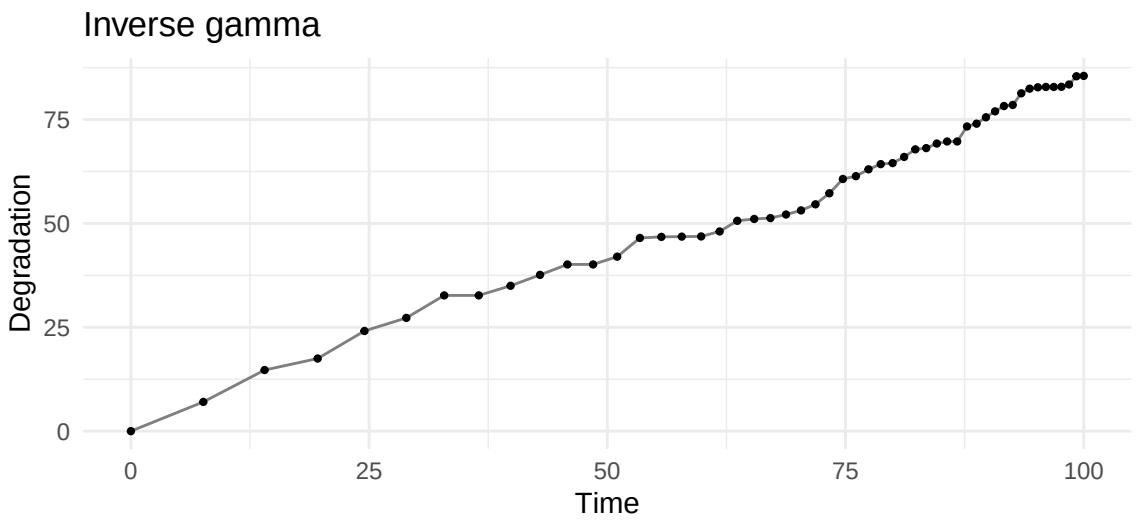
I.2.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(t2, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



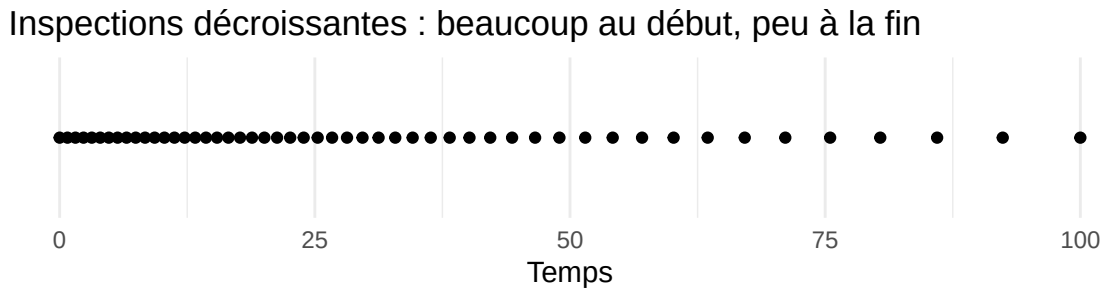
I.2.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(t2, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
Degradation
75
50
25
0
0
25
50
75
100
Time

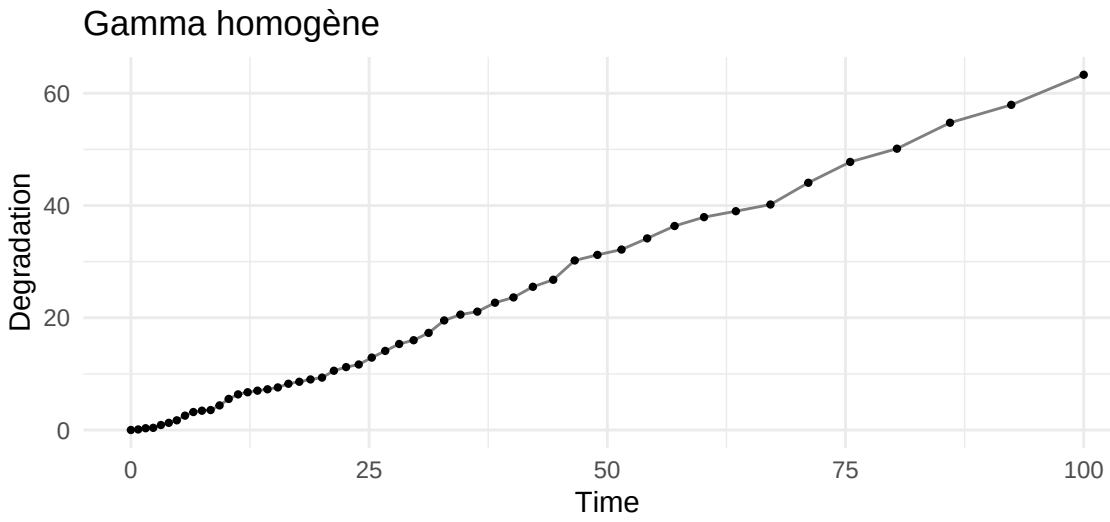
I.3 Cas 3 : inspections décroissantes

```
set.seed(3)
t3 <- inspection_times("decreasing", n = 50, tmax = 100)$time
tracer_plan(t3, "Inspections décroissantes : beaucoup au début, peu à la fin")
```



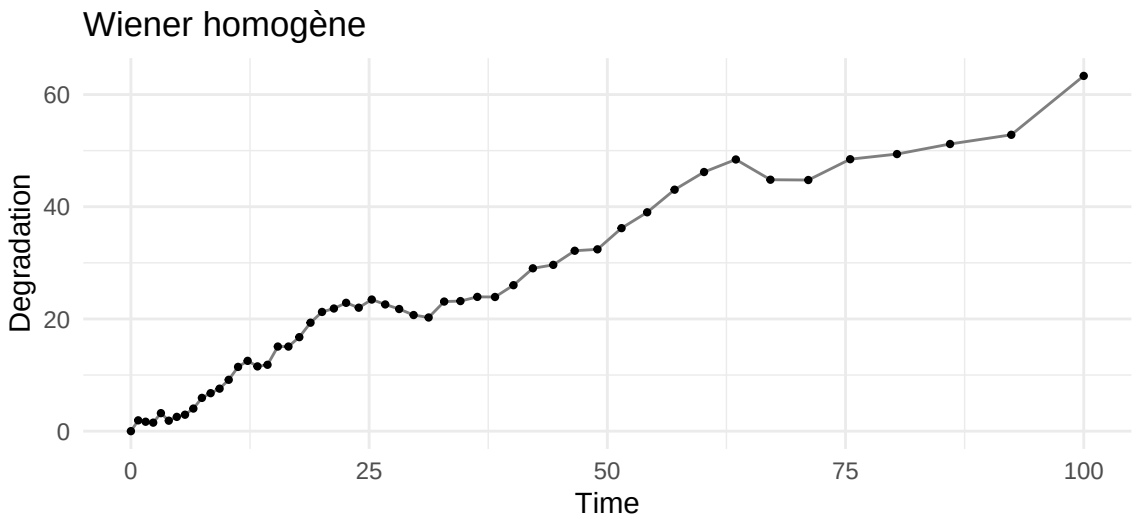
I.3.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t3, a = 2, b = 3), "Gamma homogène")
```



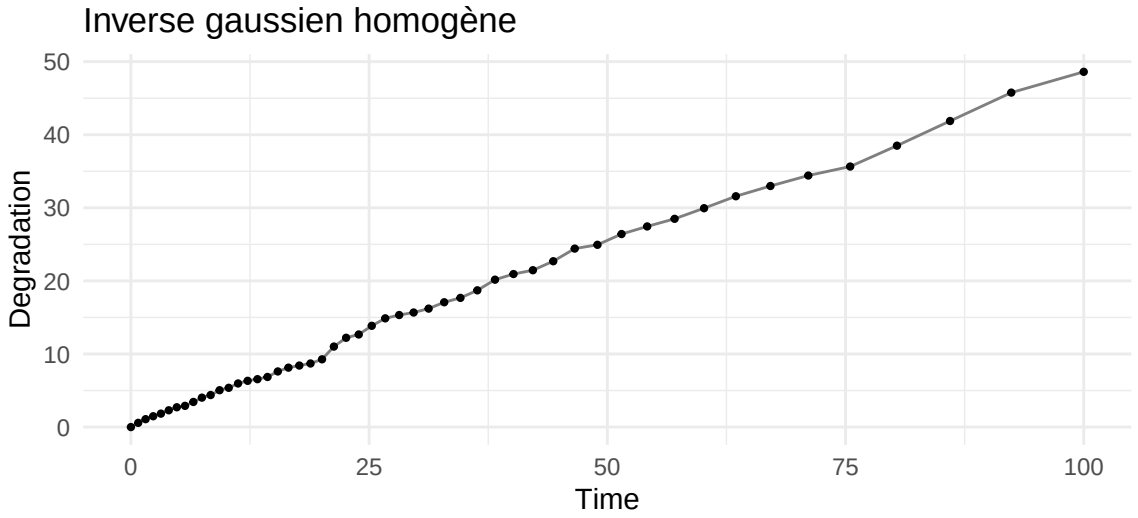
I.3.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t3, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



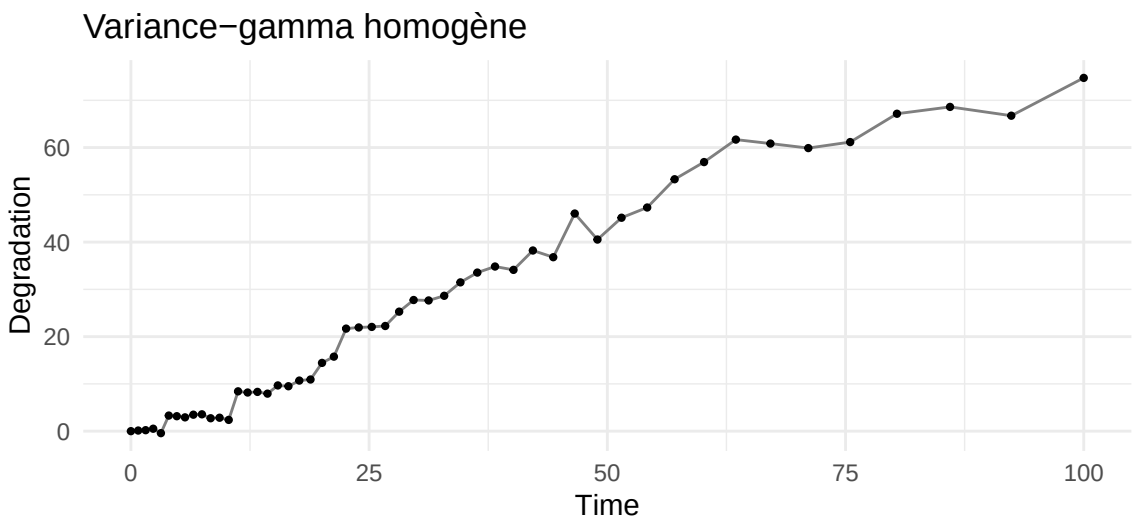
I.3.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t3, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



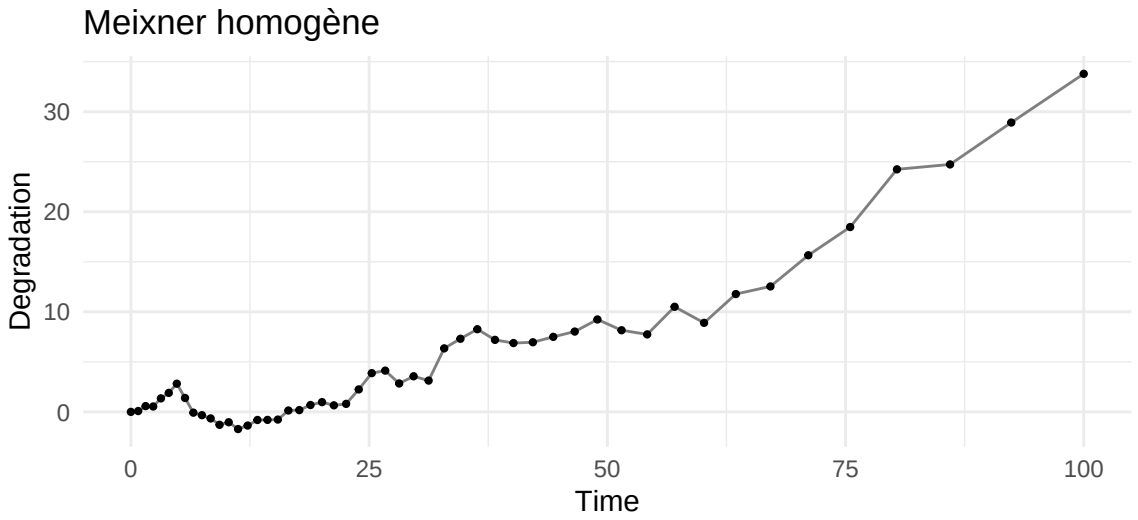
I.3.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t3, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



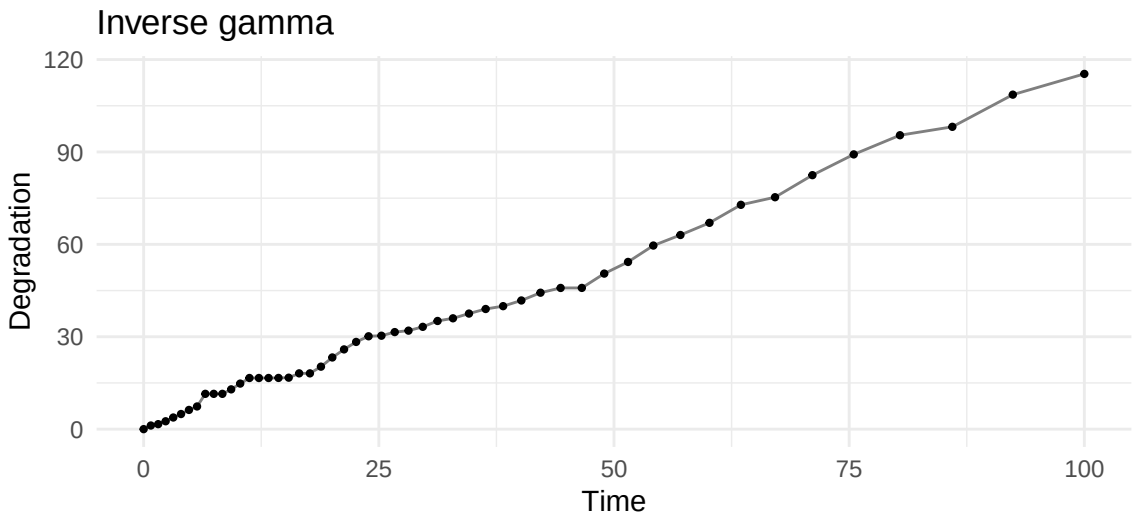
I.3.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(t3, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



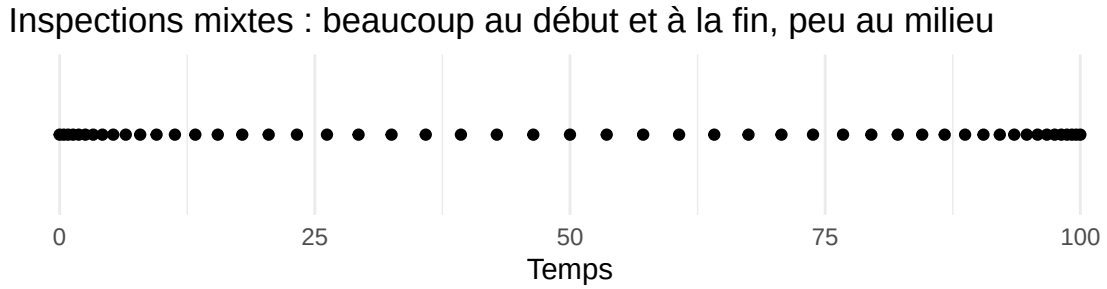
I.3.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(t3, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
120
90
60
30
0
Degradation
0
25
50
75
100
Time

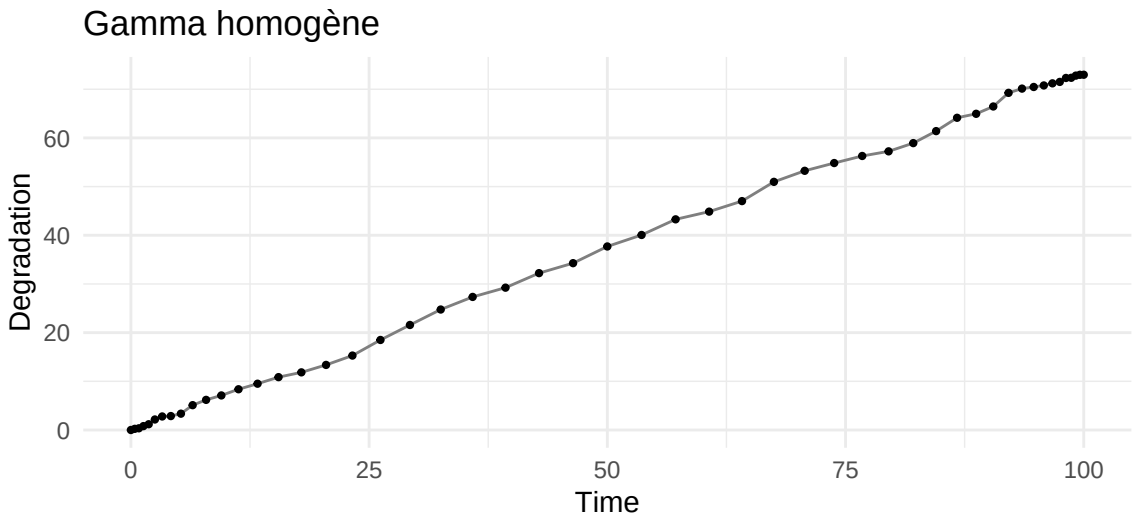
I.4 Cas 4 : inspections mixtes

```
set.seed(4)
t4 <- inspection_times("dec-inc", n = 50, tmax = 100)$time
tracer_plan(t4, "Inspections mixtes : beaucoup au début et à la fin, peu au milieu")
```



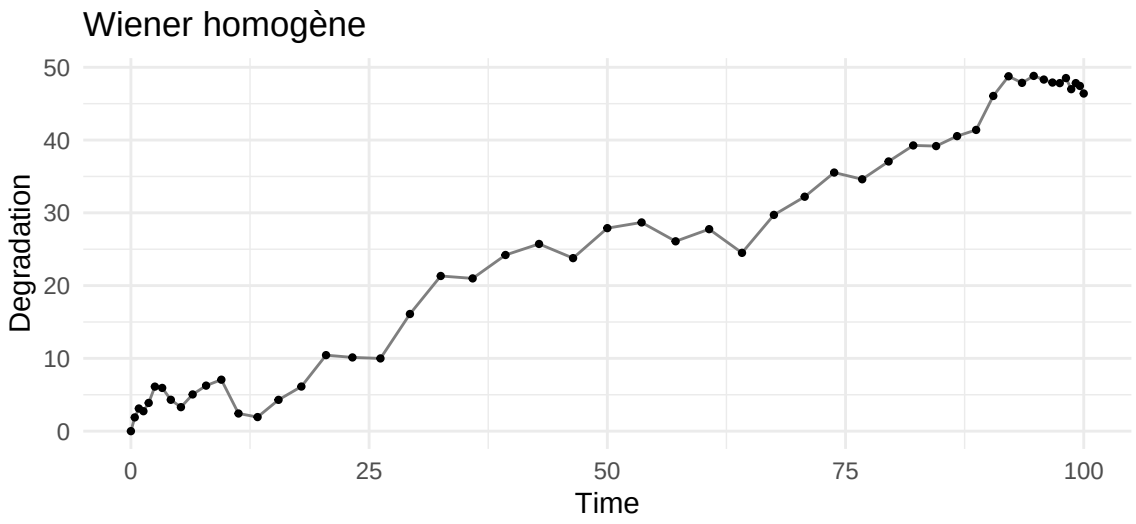
I.4.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(t4, a = 2, b = 3), "Gamma homogène")
```



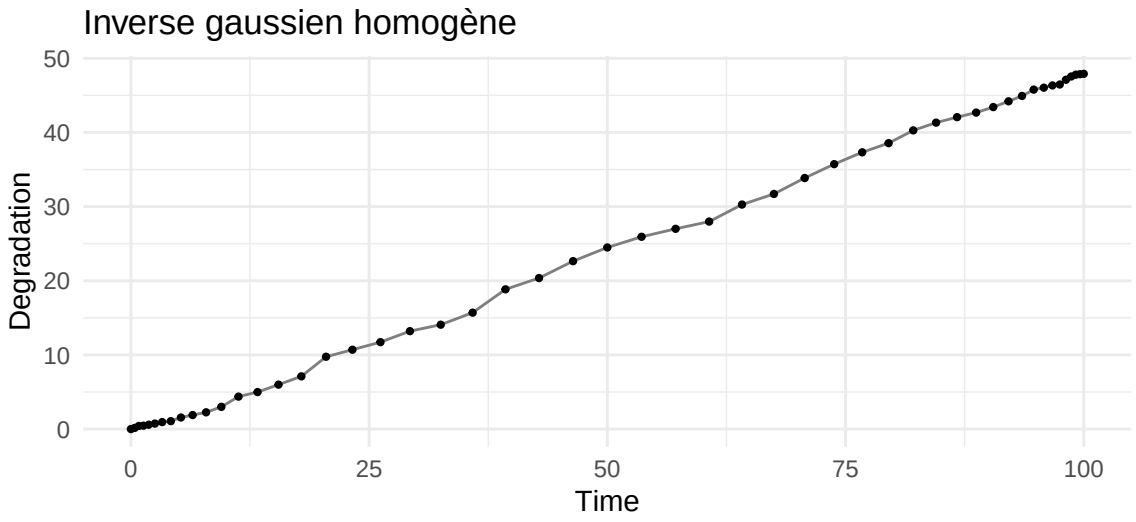
I.4.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(t4, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



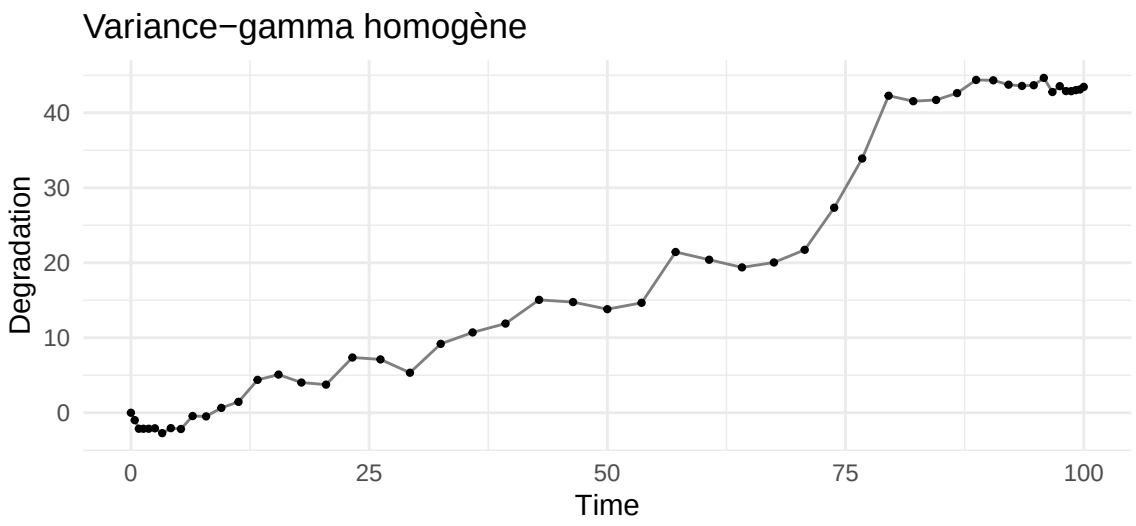
I.4.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(t4, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



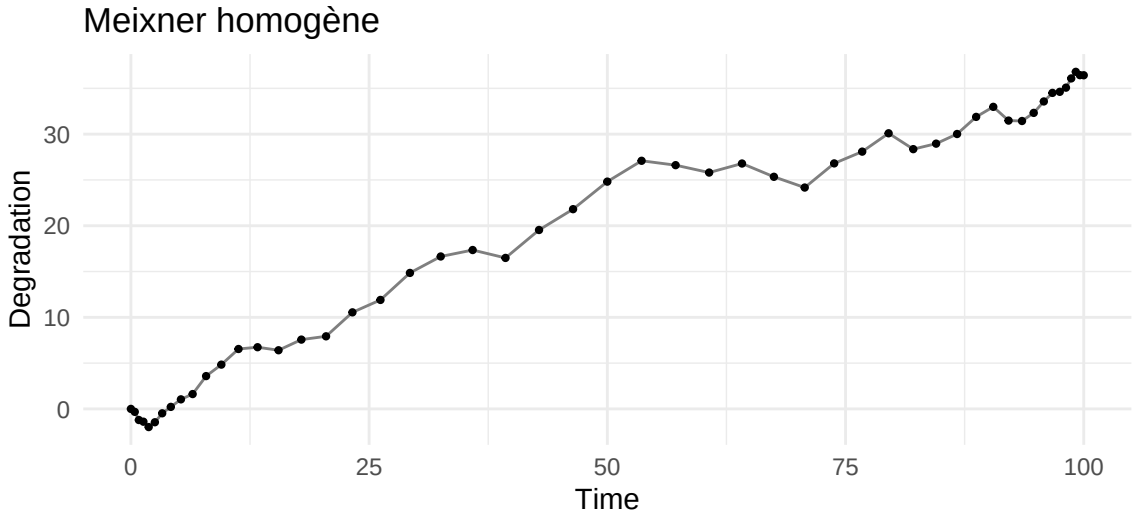
I.4.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(t4, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



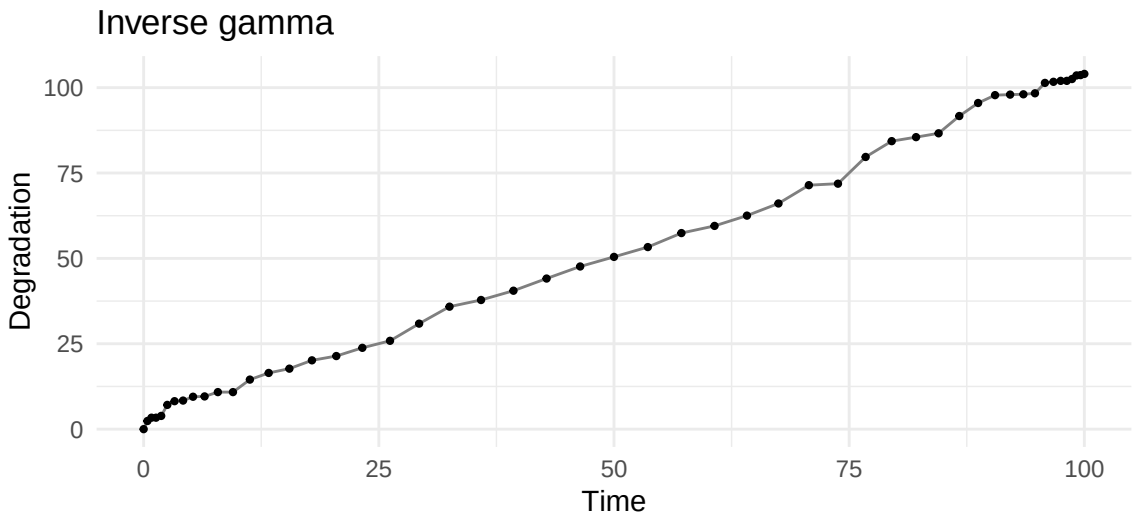
I.4.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(t4, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



I.4.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(t4, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```

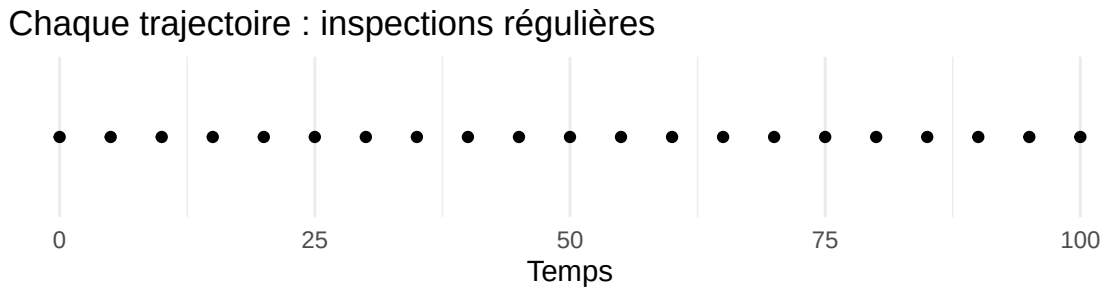

Inverse gamma
Degradation
100
75
50
25
0
0
25
50
75
100
Time

II Plusieurs trajectoires

10 trajectoires, chacune inspectée 20 fois entre 0 et 100. Le résultat de `inspection_times()` est passé directement aux fonctions `sim_` : il contient les instants de chaque trajectoire.

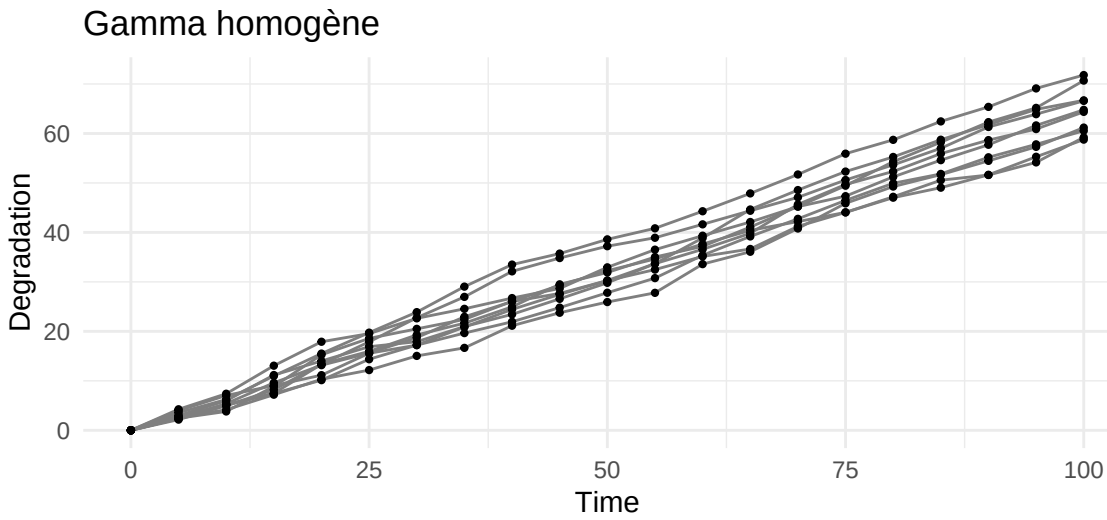
II.1 Cas 5 : inspections régulières

```
set.seed(5)
p5 <- inspection_times("regular", n = 20, n_traj = 10, tmax = 100)
tracer_plan(p5$time[p5$id == 1], "Chaque trajectoire : inspections régulières")
```



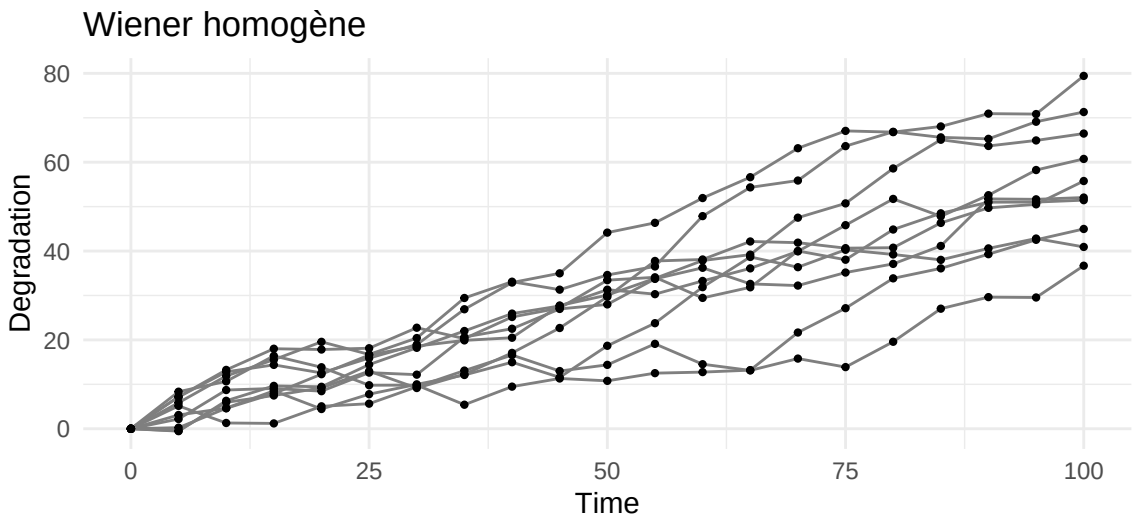
II.1.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p5, a = 2, b = 3), "Gamma homogène")
```



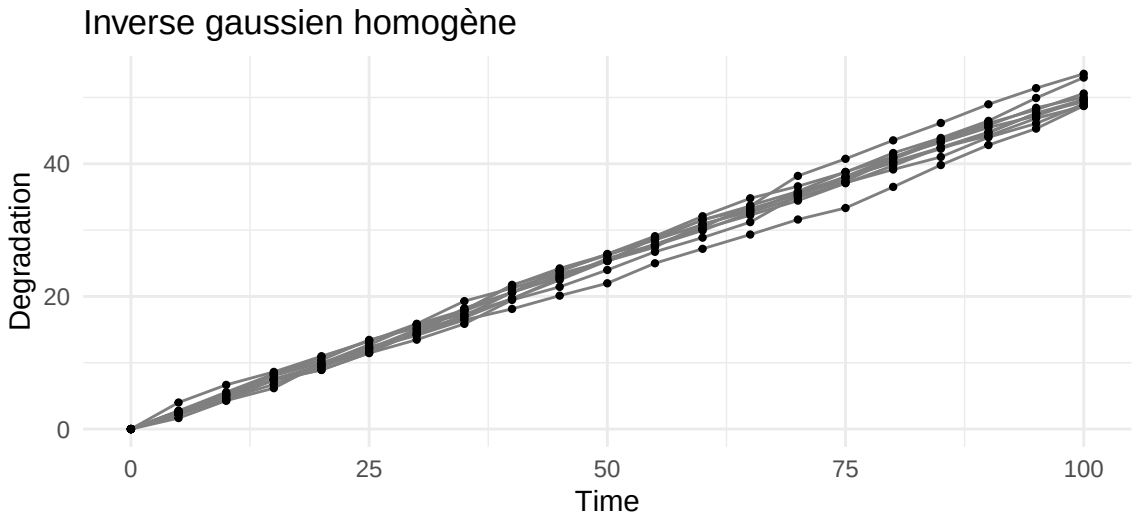
II.1.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p5, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



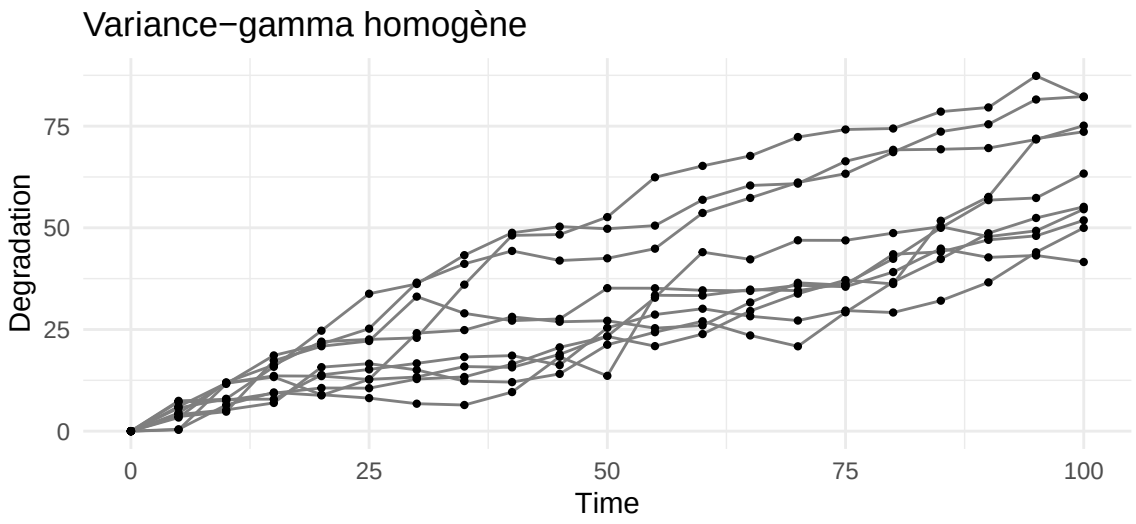
II.1.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p5, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



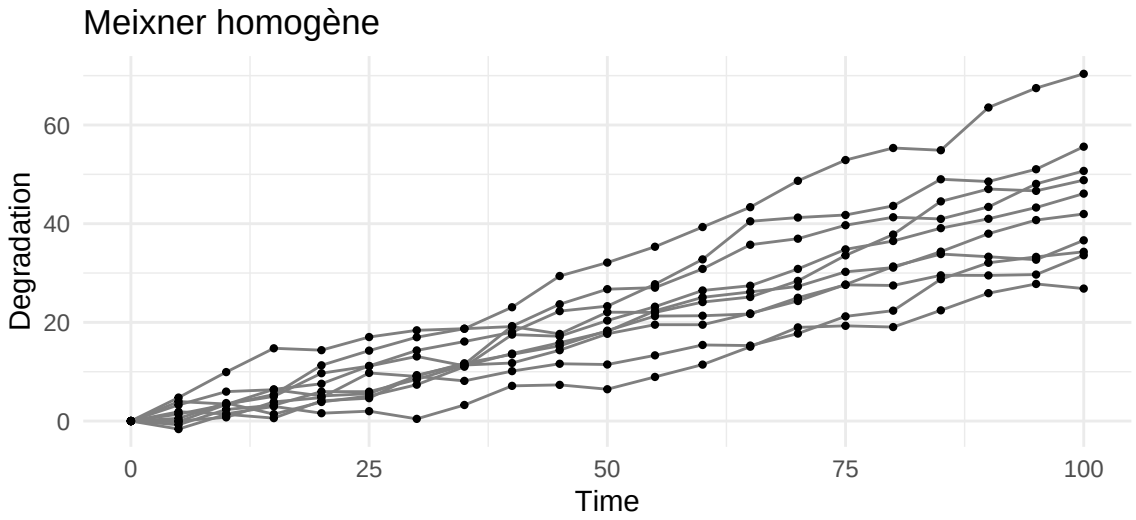
II.1.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p5, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



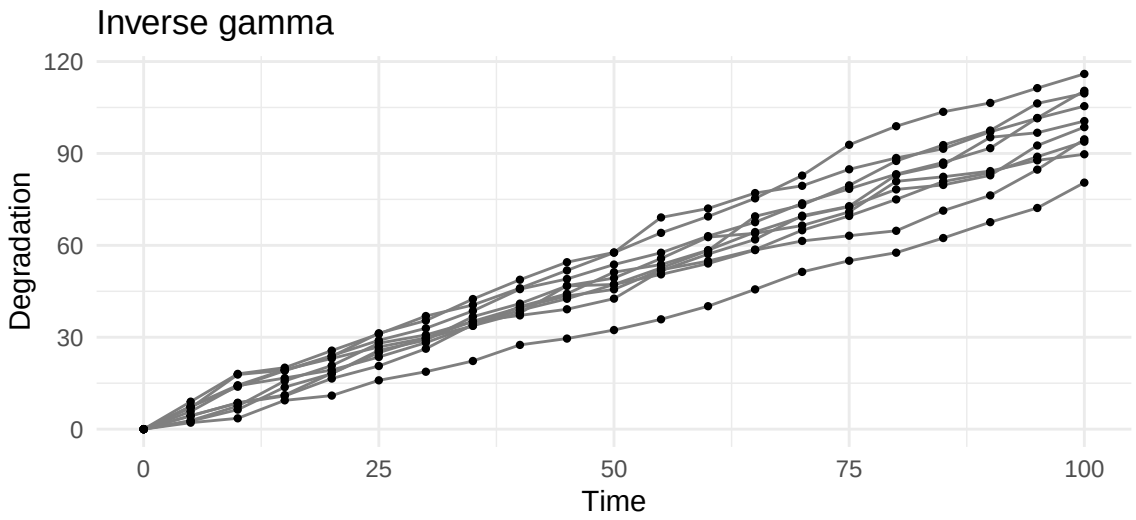
II.1.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(p5, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



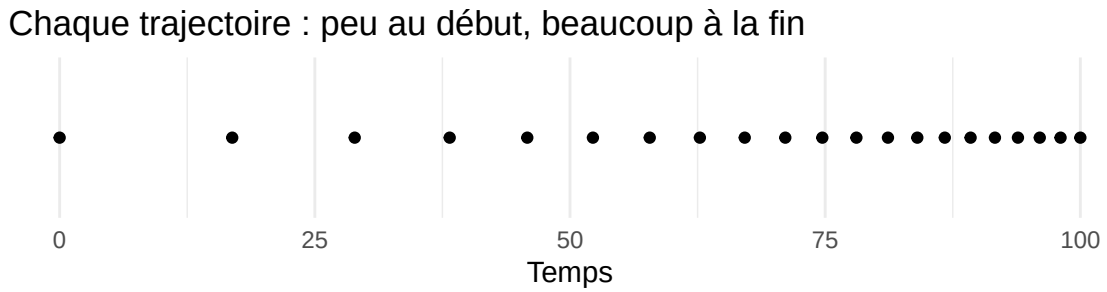
II.1.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(p5, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
120
90
60
30
0
Degradation
0
25
50
75
100
Time

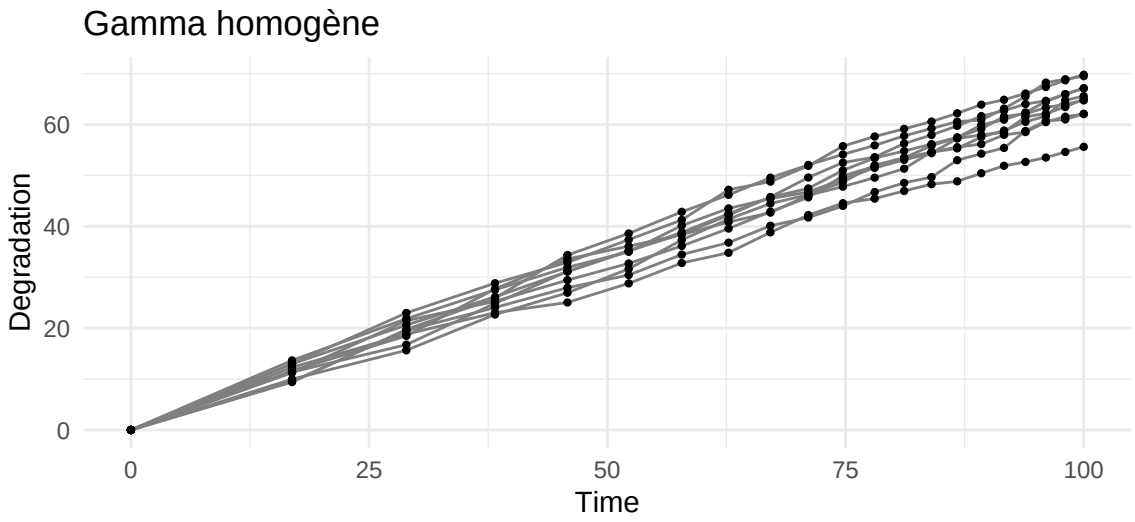
II.2 Cas 6 : inspections croissantes

```
set.seed(6)
p6 <- inspection_times("increasing", n = 20, n_traj = 10, tmax = 100)
tracer_plan(p6$time[p6$id == 1], "Chaque trajectoire : peu au début, beaucoup à la fin")
```



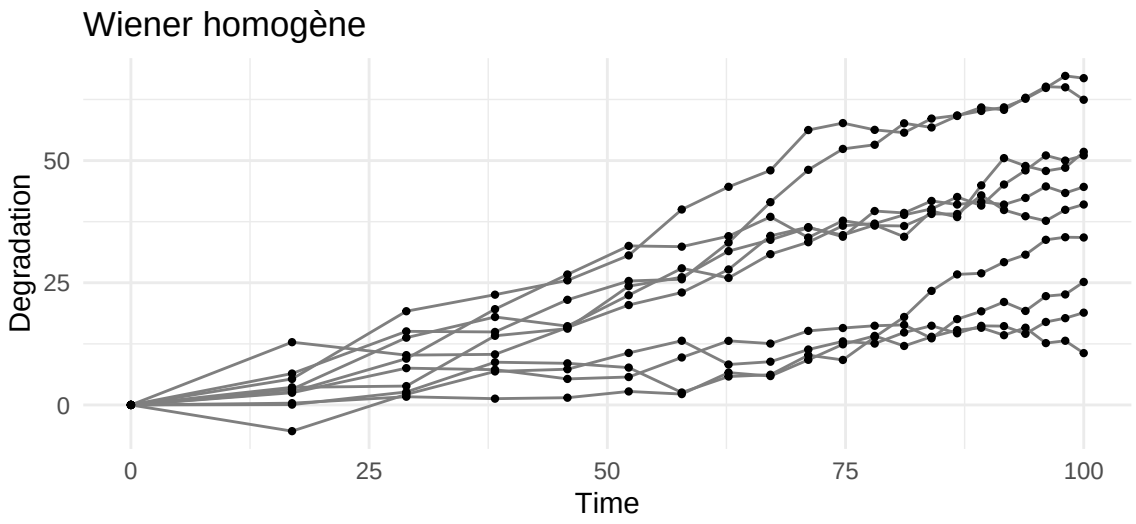
II.2.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p6, a = 2, b = 3), "Gamma homogène")
```



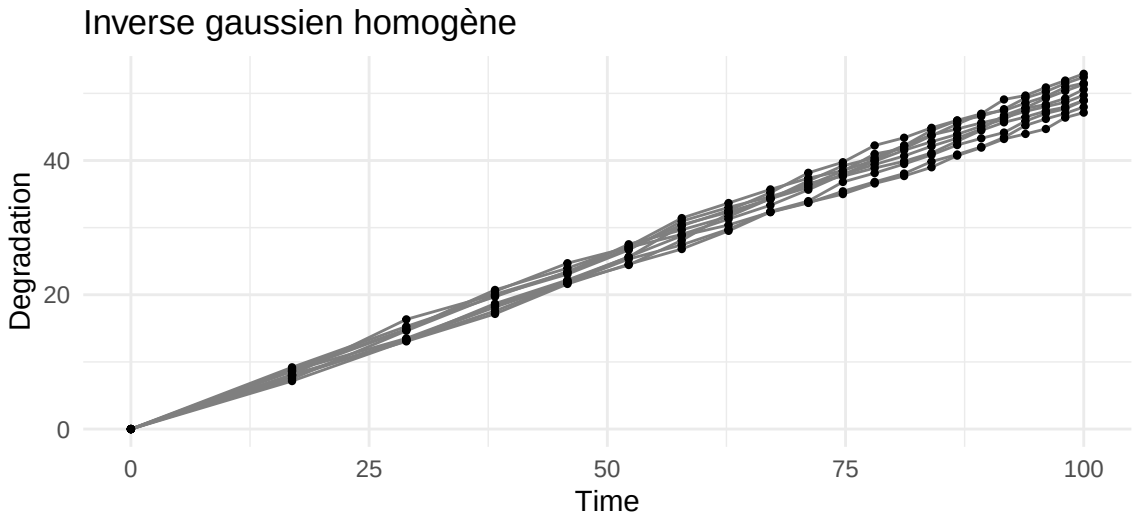
II.2.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p6, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



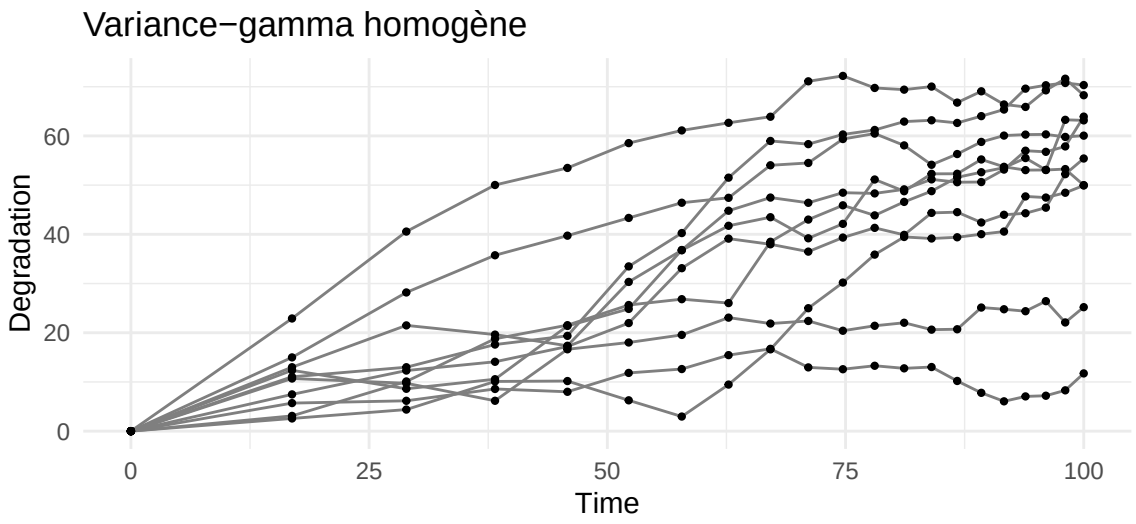
II.2.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p6, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



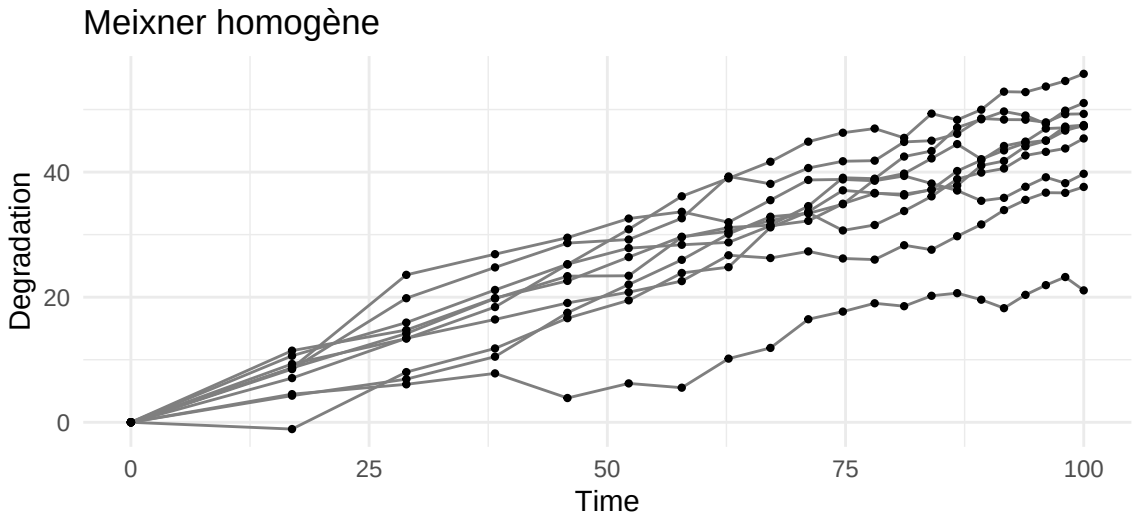
II.2.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p6, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



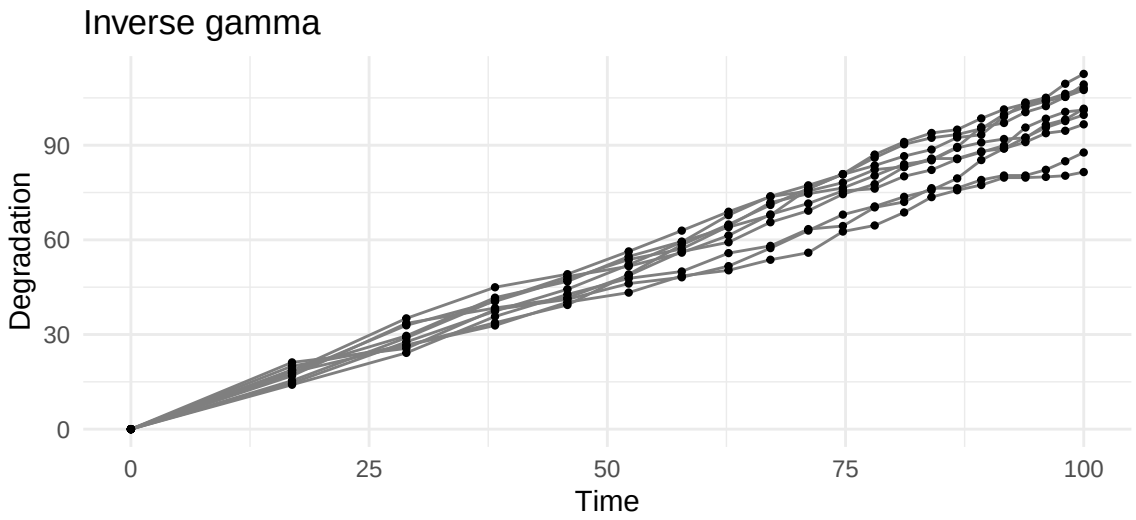
II.2.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(p6, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



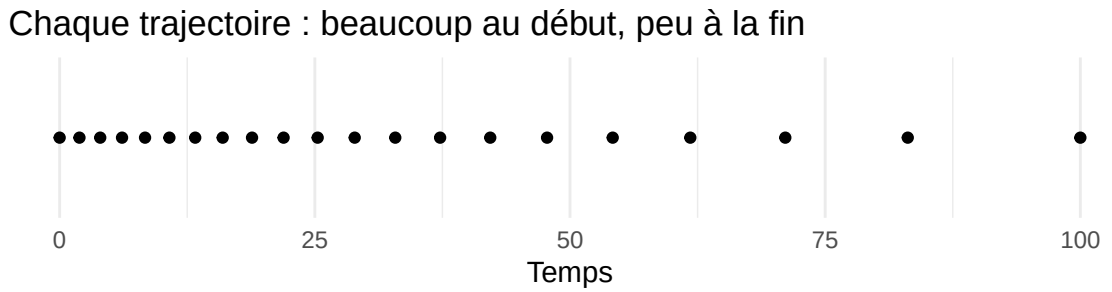
II.2.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(p6, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
Degradation
0
30
60
90
0
25
50
75
100
Time

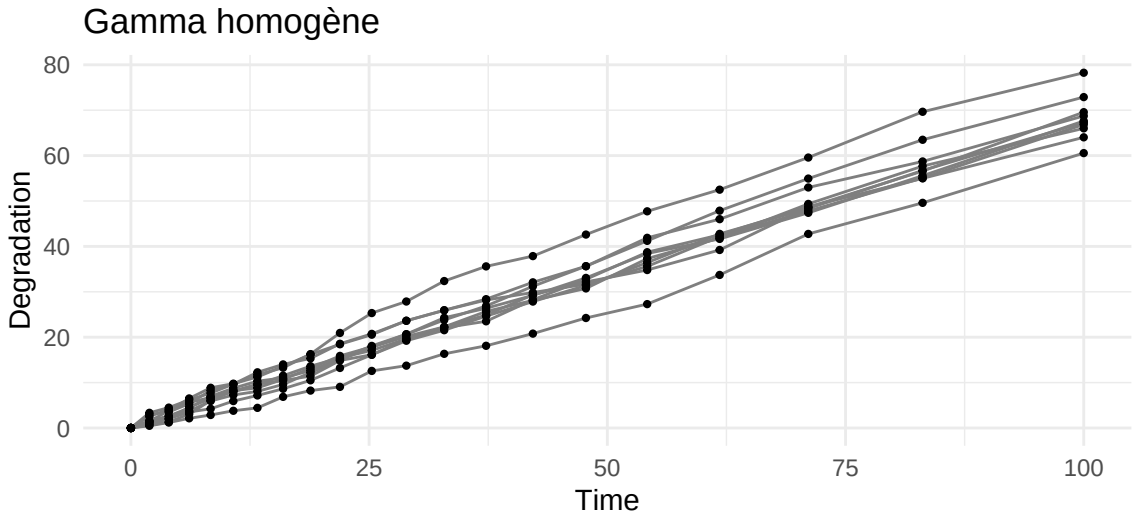
II.3 Cas 7 : inspections décroissantes

```
set.seed(7)
p7 <- inspection_times("decreasing", n = 20, n_traj = 10, tmax = 100)
tracer_plan(p7$time[p7$id == 1], "Chaque trajectoire : beaucoup au début, peu à la fin")
```



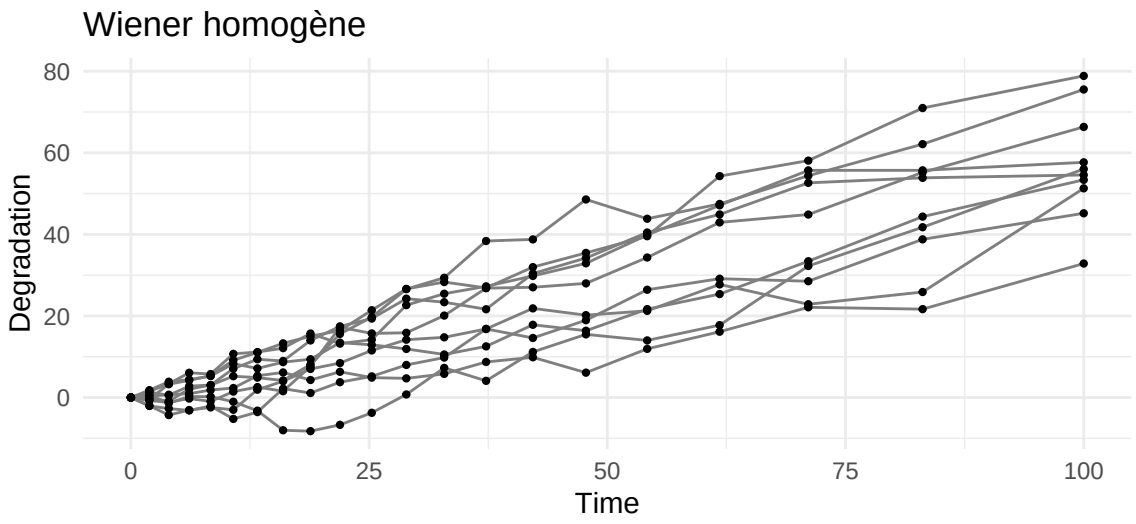
II.3.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p7, a = 2, b = 3), "Gamma homogène")
```



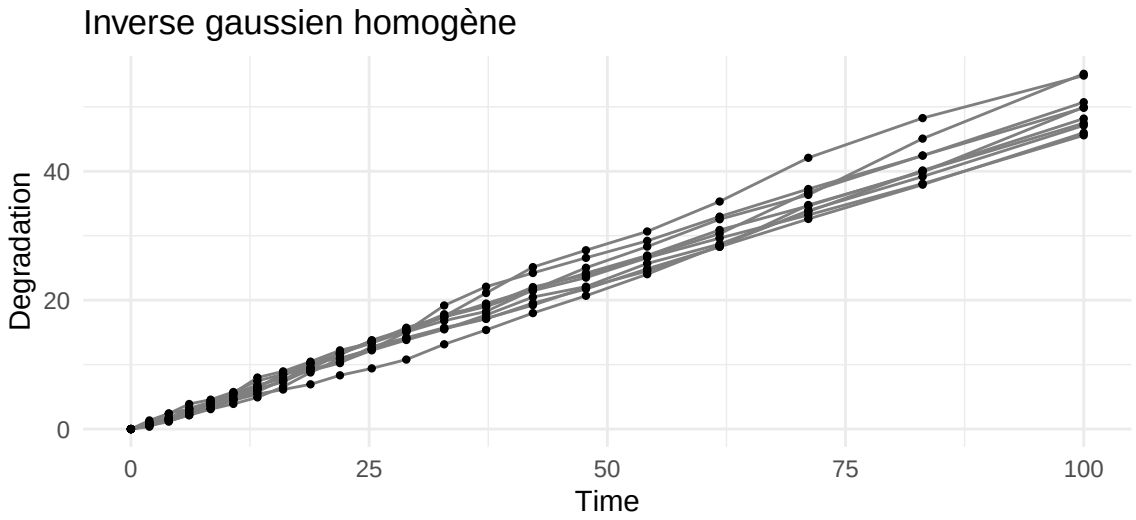
II.3.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p7, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



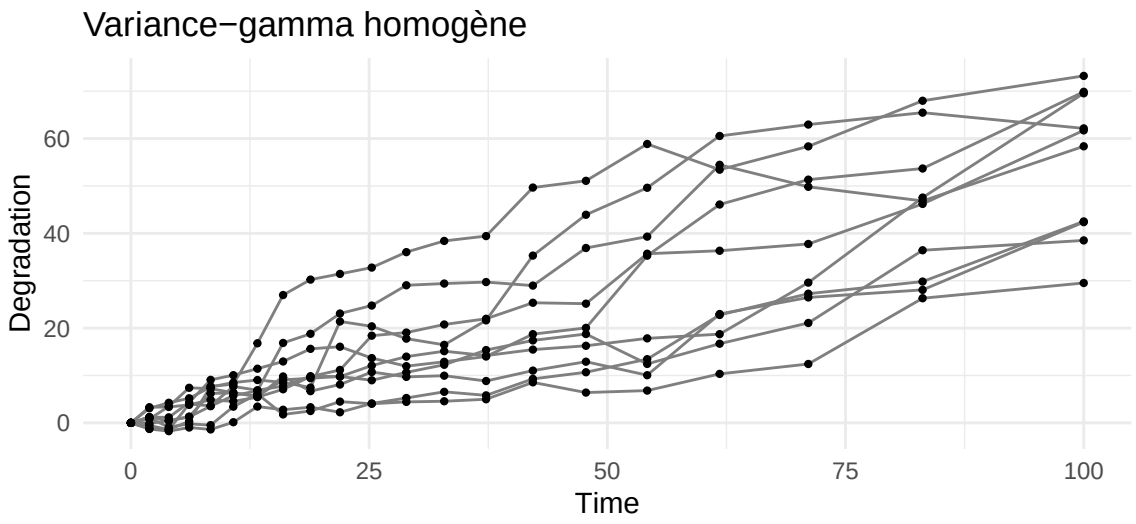
II.3.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p7, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



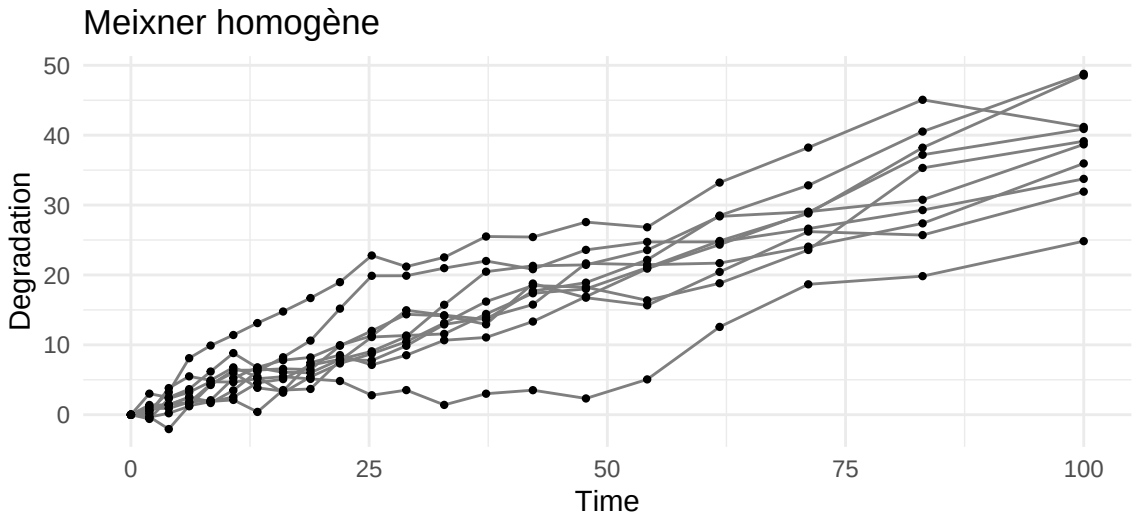
II.3.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p7, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



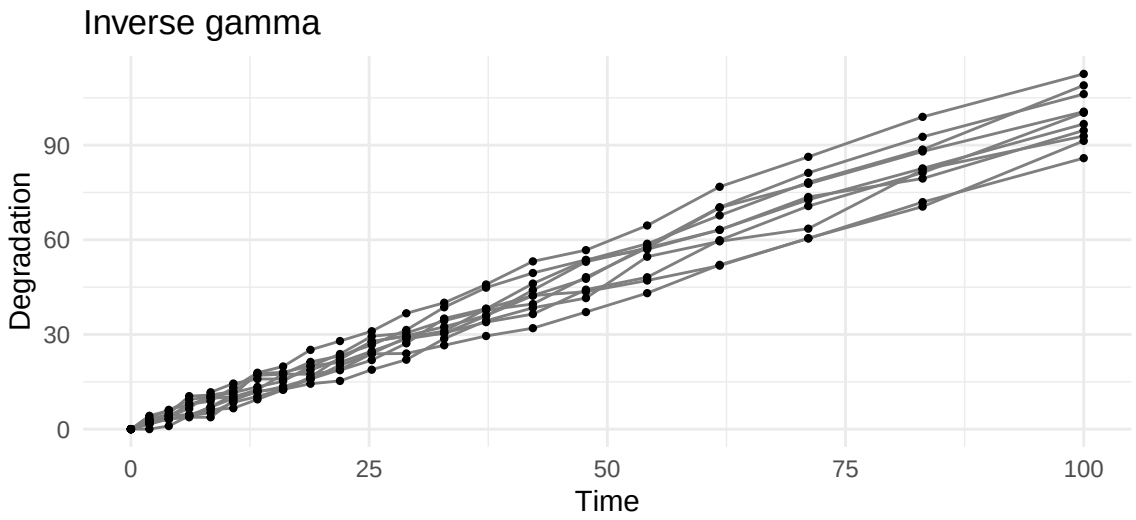
II.3.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(p7, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



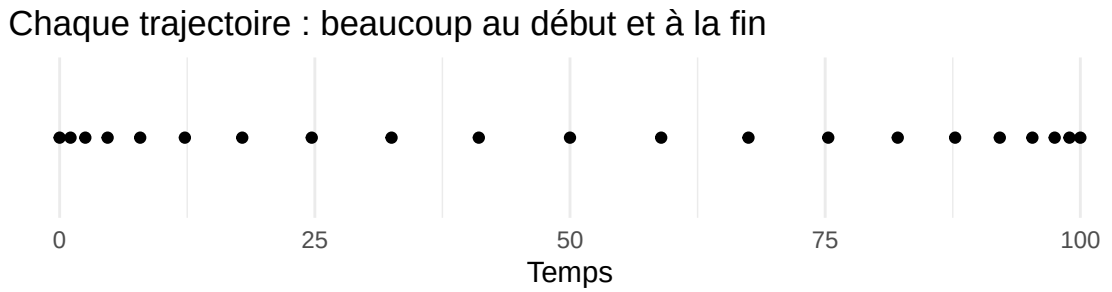
II.3.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(p7, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
Degradation
90
60
30
0
0
25
50
75
100
Time

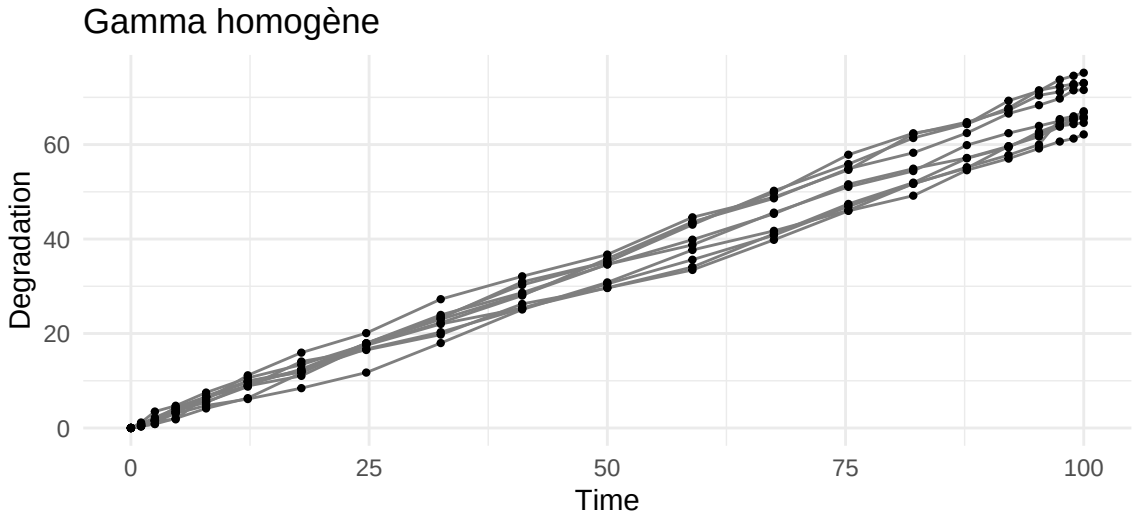
II.4 Cas 8 : inspections mixtes

```
set.seed(8)
p8 <- inspection_times("dec-inc", n = 20, n_traj = 10, tmax = 100)
tracer_plan(p8$time[p8$id == 1], "Chaque trajectoire : beaucoup au début et à la fin")
```



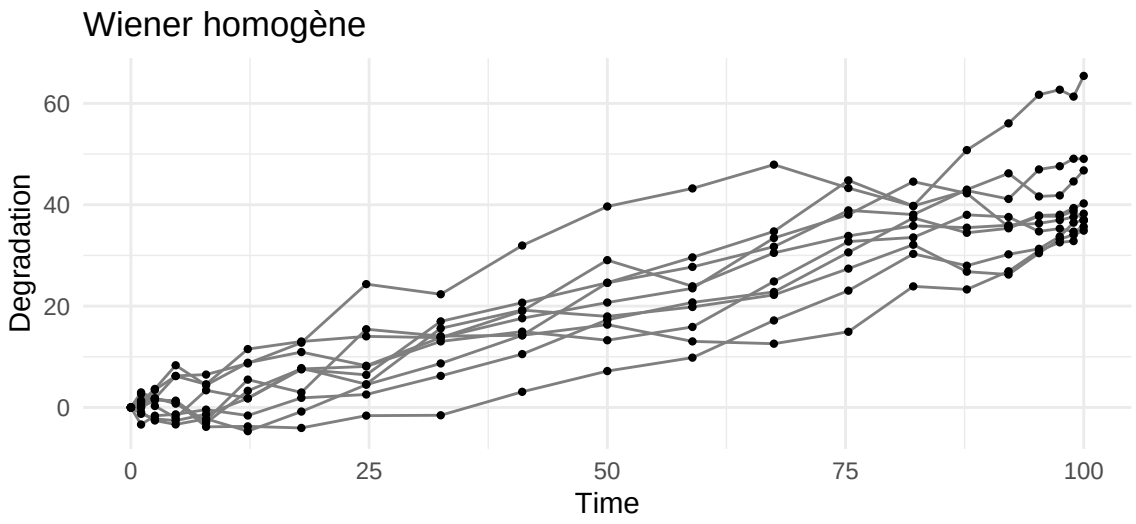
II.4.1 Modèle 1 : gamma homogène

```
plot_traj(sim_gamma(p8, a = 2, b = 3), "Gamma homogène")
```



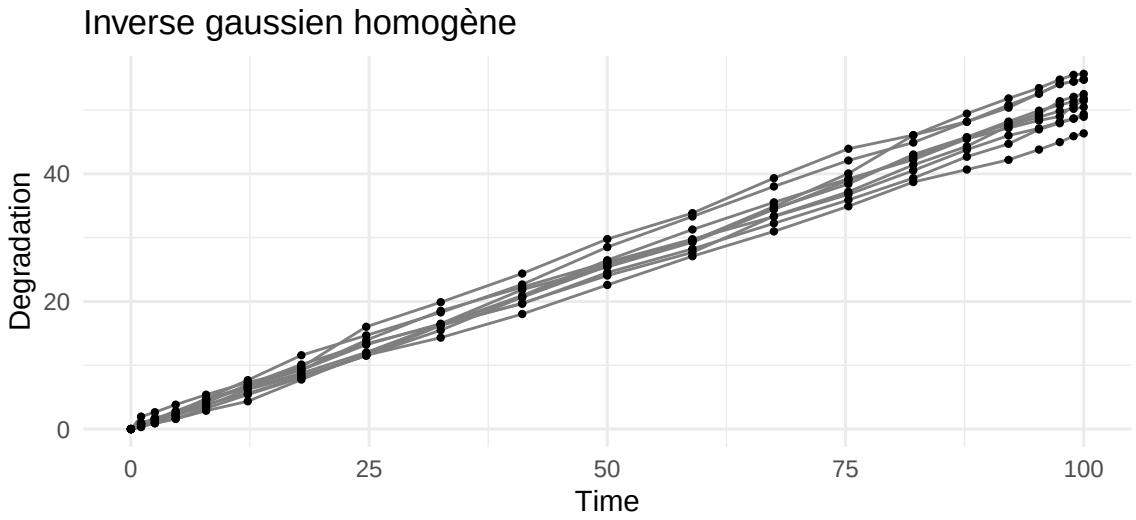
II.4.2 Modèle 2 : Wiener homogène

```
plot_traj(sim_wiener(p8, mu = 0.5, sigma = 1.5), "Wiener homogène")
```



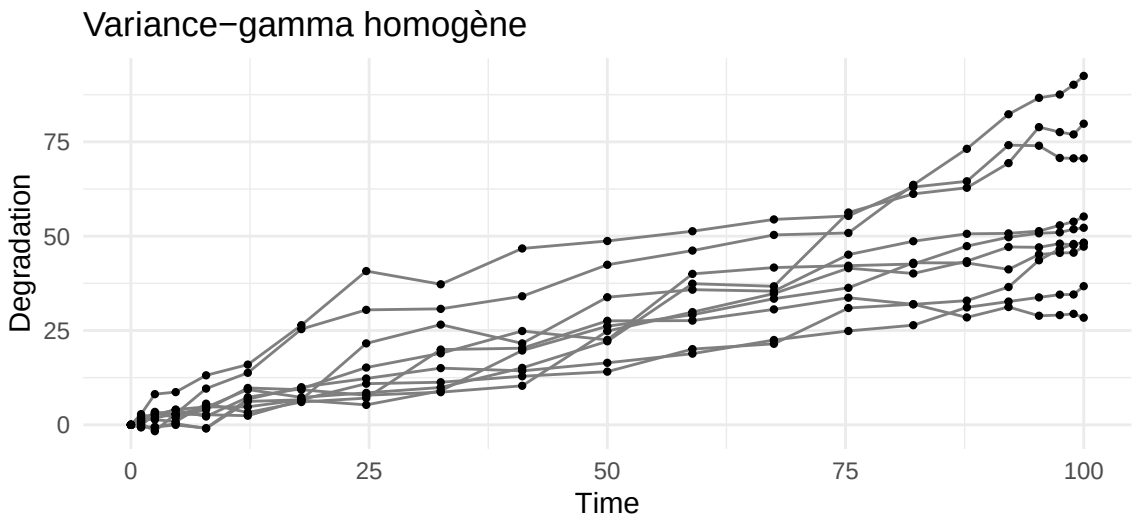
II.4.3 Modèle 3 : inverse gaussien

```
plot_traj(sim_ig(p8, mu = 0.5, eta = 2), "Inverse gaussien homogène")
```



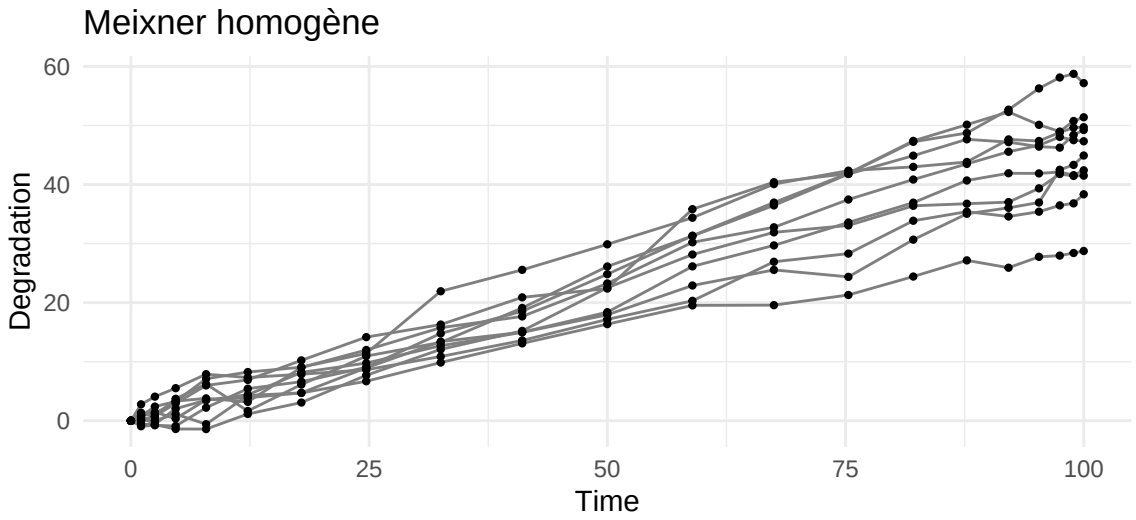
II.4.4 Modèle 4 : variance-gamma

```
plot_traj(sim_vg(p8, theta = 0.5, sigma = 1.5, nu = 2), "Variance-gamma homogène")
```



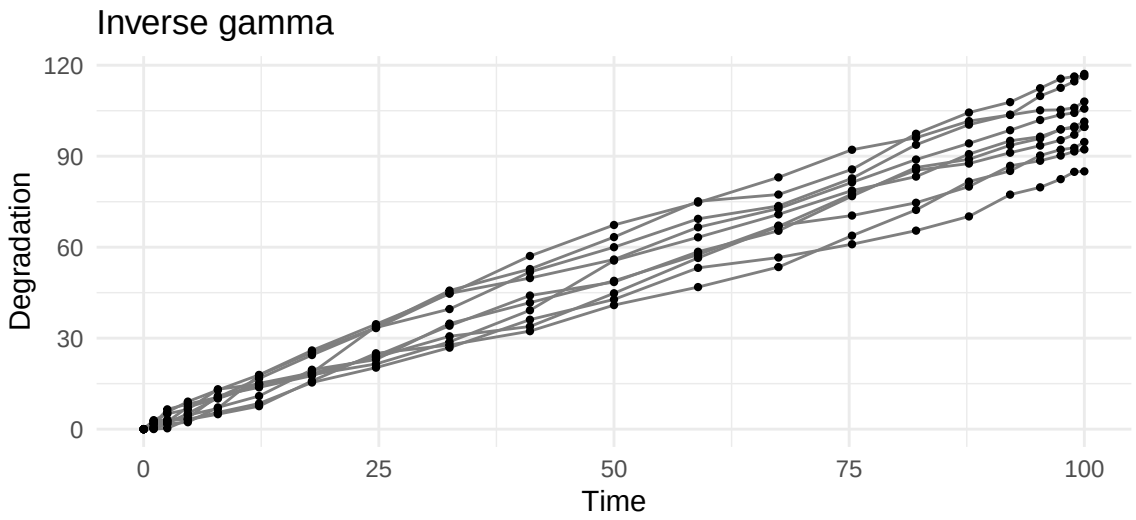
II.4.5 Modèle 5 : Meixner

```
plot_traj(sim_meixner(p8, a = 1, b = 0.3, d = 1.5, m = 0.2), "Meixner homogène")
```



II.4.6 Modèle 6 : inverse gamma

```
plot_traj(sim_igam(p8, theta = 1, alpha = 1, beta = 1), "Inverse gamma")
```


Inverse gamma
Degradation
120
90
60
30
0
0
25
50
75
100
Time

III Récupérer les données simulées

Le résultat d'une fonction `sim_` s'affiche directement : modèle, paramètres et premières valeurs.

```
set.seed(9)
plan <- inspection_times("increasing", n = 10, n_traj = 3, tmax = 100)
x <- sim_gamma(plan, a = 2, b = 3)
x
```

```
## 3 simulated trajectory(ies) of the gamma process
## parameters: a = 2, b = 3
##
##   id        t        X
## 1  1  0.00000  0.00000
## 2  1 28.90648 17.21830
## 3  1 45.81569 26.86238
## 4  1 57.81297 34.25179
## 5  1 67.11877 44.19902
## 6  1 74.72217 50.52252
```

`x[[k]]` donne la k-ième trajectoire, sous la forme d'un `data.frame(t, X)` :

```
x[[2]]
```

```
##            t        X
## 1    0.00000  0.00000
## 2   28.90648 22.48570
## 3   45.81569 35.94921
## 4   57.81297 46.52566
## 5   67.11877 53.58187
## 6   74.72217 59.26301
## 7   81.15076 62.37490
## 8   86.71945 66.99940
## 9   91.63138 69.44171
## 10  96.02526 72.29244
## 11 100.00000 76.05928
```

`traj_df()` rassemble toutes les trajectoires dans un seul tableau, avec une colonne `id` :

```
d <- traj_df(x)
head(d)
```

```
##   id        t        X
## 1  1  0.00000  0.00000
## 2  1 28.90648 17.21830
## 3  1 45.81569 26.86238
## 4  1 57.81297 34.25179
## 5  1 67.11877 44.19902
## 6  1 74.72217 50.52252
```

```
d[d$t == 100, ]
```

```
##    id   t        X
## 11  1 100 66.57627
## 22  2 100 76.05928
## 33  3 100 66.18547
```

Ce tableau peut être enregistré, par exemple dans un fichier CSV :

```r
write.csv(d, "trajectoires_gamma.csv", row.names = FALSE)
```

Le graphique renvoyé par `plot_traj()` se modifie avec les fonctions de `ggplot2`, par exemple pour ajouter un seuil de défaillance :

```r
plot_traj(x, "Gamma homogène : 3 trajectoires") +
  geom_hline(yintercept = 60, linetype = "dashed", colour = "red") +
  labs(x = "Temps (heures)", y = "Dégradation")
```

